

From Son to State, from State to Son: China's Rural Pension across Three Generations

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July 2026

Abstract

Old-age support in rural China was long the private obligation of adult children; in 2016 a unified national pension gave it a public counterpart, a monthly benefit at sixty regardless of past contributions. Using the age-sixty eligibility cutoff in a difference-in-differences design across three national surveys, we trace the pension through three generations of rural households. Pension receipt raises grandparents' care of grandchildren by eight percentage points. Parents, in turn, stay away longer: children spend 1.4 fewer months a year with their fathers and 1.6 fewer with their mothers. More children receive homework help from grandparents, and conflict between parents and children becomes more frequent; both changes are concentrated among boys. What the household gained economically, the grandchild partly paid in childhood.

JEL No. I38, J13, J14

Keywords: rural pension, migration, intergenerational transfers, China

1 Introduction

Much depends on how the family's economic means are used to cater to the interests of different individuals in the household: women and men, girls and boys, children and adults, old and young.

Amartya Sen, *Development as Freedom*, p. 193

In 2016, for the first time, a unified national pension reached the Chinese countryside: rural residents turning sixty could draw a monthly benefit, whether or not they had ever contributed. Support in old age, long the duty of adult children, acquired a public substitute. The timing mattered, because rural parents faced a trade between earnings and presence: work outside agriculture paid about one third more than farm work for the same worker (Gai et al., 2025), and much of it lay far from home. By 2015, 169 million rural workers were employed away from their home townships (National Bureau of Statistics of China, 2016), leaving an estimated 61 million children behind in the villages (Duan et al., 2013). For many of the parents who stayed, what bound them was not the field but two claims at home: aging parents and young children. The pension could ease only the first. We ask what became of the second: who left, who stayed, and who took over the care of young children. Our answer is that the pension kept parents away longer, moved grandparents into the vacancy, and charged part of the cost to the grandchild.

The New Rural Pension Scheme opened in a tenth of China's counties in 2009 and covered them all by 2012; in 2014 it was merged with its urban counterpart, and by 2016 the consolidation was complete (Huang and Zhang, 2021). Anyone past sixty drew the benefit without ever having paid in. The floor was 55 yuan a month, about an eighth of rural income per head (National Bureau of Statistics of China, 2010). The benefit was small, but it pressed on the oldest arrangement in the countryside: *yang'er fanglao*, rearing a son to insure one's old age. A pension that assumes part of a son's duty frees part of the son's time, and the time he withdraws must be filled by someone. In rural China that someone is a grandparent, and the substitution is not neutral: children in grandparents' care fall behind in school, most where the grandparents are least schooled, and most among boys (Wang and Bansak, 2024).

We follow the pension through the household: from the grandparent who draws it, through the adult child it releases from a monthly duty, to the grandchild whose care it rearranges. The rule that assigns the benefit provides the test. Eligibility begins at sixty, so within the same village two grandparents a year apart in age stand on opposite sides of the line, and the 2016 consolidation drew that line nationwide. We compare households just above and just below sixty, before and after 2016, in three national surveys: the China Health and Retirement Longitudinal Study (CHARLS), which records what the elderly receive and what their children send; the China Family Panel Studies (CFPS), which records whom children live with and who helps with their homework; and the China

Migrants Dynamic Survey (CMDS), which records the families that migrate. Together they show each generation's side of the same exchange.

Three main facts emerge from these data. The grandparents took the benefit: at sixty, after 2016, pension receipt jumps by eighteen percentage points; the money the elderly send out to their adult children falls by nearly two-fifths, and it falls only in households with an adult child to send it to; the share of grandparents caring for a grandchild rises by eight percentage points, nearly ten among grandmothers. The parents stayed away longer: children spend 1.4 fewer months a year living with their fathers and 1.6 fewer with their mothers, yet their parents are no likelier to work outside the county or the province, or to be absent from home altogether. And inside the household one task changed hands: homework. More children get help with it from a grandparent, quarrels between parents and children grow more frequent, and both changes fall almost entirely on boys.

The three facts are consistent with a single mechanism. The pension did not work through the household's budget: the benefit came to an eighth of rural income, and the money grandparents sent to their adult children fell rather than rose. It worked through the household's obligations. A grandparent with a pension needed less from an adult child: less money, and fewer visits home. The adult children who already worked away stayed away longer; those who had stayed home did not now leave. And the grandmother, needing less, gave more: she took over the grandchild's days, homework included. This is why the literature disagrees about whether the pension moved anyone: Gai et al. (2025) find that it freed adult children to leave the farm, while Huang and Zhang (2021) counted migrants and found no effect. Both are right. The pension did not change how many people left; it changed how long they stayed away.

What the design adds is a view inside the child's day. That an old-age pension can reach a grandchild is already known: in South Africa a generous pension paid to grandmothers raised the height and weight of the granddaughters who lived with them (Duflo, 2003). That a grandparent schooled a generation earlier makes an imperfect substitute for a parent is known too (Wang and Bansak, 2024), and the literature on China's left-behind children has measured the gaps they carry in schooling and health. Those studies count outcomes. The eligibility cutoff lets us count inputs: which of a child's ordinary tasks passes from parent to grandparent when old-age support turns public. Most stay where they were. The night-time care, the school run, the bedtime: whoever did them before the pension still does them after. One task changes hands. As the parents are seen less,

the homework passes to the grandparent. Quarrels with the parents grow more frequent. And the changes settle on the sons.

It also changes the accounting. By the ordinary measure the reform succeeds: it raised the incomes of the old and, by freeing their children for more productive work, lifted rural income in the aggregate (Gai et al., 2025). That measure adds up the members of the household who chose: the grandparent who drew the benefit and the parent who earned the wage. The child made neither decision, and no one asked them. Their cost is not one of income but of capability (Sen, 2001): a parent seen less, homework handed to a grandparent with fewer years of school, more of their day spent with the generation the pension was meant to relieve. Counted as one household, the reform comes out ahead. Counted by member, one member does not.

2 Institutional Background

2.1 The *Hukou* and the New Rural Pension Scheme

The household this paper studies sits between two institutions. The older is the household registration, or *hukou*, which since the 1950s has tied a person's schooling, health care, and relief to their place of origin; a rural parent who takes work in a city rarely brings the family along. The newer is the New Rural Pension Scheme, which by 2016 had reached rural residents nationwide for the first time. The first institution settled where the children would be; the second gave the old public support where there had once been only their sons.

2.2 Old-Age Support and the Pension

Provision for the old had never been the state's affair in the countryside. Under the social norm of *yang'er fanglao*, a household reared sons partly as insurance, and the insurance was paid inside the family: money sent home, and the presence that care demanded. Rapid growth in the 2000s made the arrangement fragile, because the growth happened elsewhere. Towns and cities paid adult children more than the village could, and the cash that supported the family was increasingly earned away from it. When these workers left, two questions followed: who would support their parents, and who would raise their children.

The New Rural Pension Scheme answered the first question. It started in about a tenth of China's

counties in 2009 and covered all counties by 2012 (Huang and Zhang, 2021). The scheme had two parts. A rural resident could pay into an individual account, at a level of their own choosing. On top of the account, everyone who reached sixty received a flat basic pension of 55 yuan a month, paid from the public budget (Huang and Zhang, 2021). In 2014 the rural scheme was merged with the urban residents' scheme into one program, the Urban and Rural Resident Pension. By 2016 the merged program operated nationwide. This is why we study 2016: at the end of that year the program had 508 million participants, and 153 million of them were receiving benefits (Ministry of Human Resources and Social Security of the People's Republic of China, 2017). Most recipients were rural. For a rural resident past sixty, nearly every rural resident past sixty was receiving one.

Eligibility for the basic pension did not depend on a contribution record. A person who was already sixty when the New Rural Pension Scheme reached their county began to draw the flat pension at once, whether or not they had ever paid a premium (Huang and Zhang, 2021). But eligibility had a second condition, and it fell on the family, not on the elderly person alone. In many places, a parent could draw the basic pension only after at least one adult child enrolled in the scheme and began to pay (Chen et al., 2020). What mattered was not the parent's own record, but the child's enrollment. The sum itself was small: fifty-five yuan a month, the floor set nationwide at the program's start (Huang and Zhang, 2021).

A separate literature asks what a pension does inside the family once it is paid. One study finds that a parent's own pension crowds out support from children: as the pension rises, adult children transfer less money to their parents (Nikolov and Adelman, 2019). Whether the pension also frees working-age children to leave the farm is contested: one study finds that it does (Gai et al., 2025); another finds only a shift from farmwork to nonfarmwork (Huang and Zhang, 2021). The study closest to this one follows the pension further into the family: Guo et al. (2025) find that it lowers a son's eldercare burden, lowers the bride price his family pays, and moves the sex ratio at birth toward balance. That literature reaches the parent's decisions about marriage and children. In rural China, 43.9 percent of parents past sixty lived with an adult child, and another 25.6 percent lived in the same village (Chen et al., 2021). The literature stops at the parent. It never reaches the grandchild who was already there.

2.3 Migration, the *Hukou*, and *Left-Behind* Childhood

A child's *hukou* stays in the village even after a parent leaves for work, and a village *hukou* does not entitle a child to a place in a city public school. Tens of millions of rural parents have chosen, for this reason among others, to leave their children behind rather than bring them along (Duan et al., 2013). Those children have been studied for two decades. Growing up without both parents present lowers a child's cognitive test scores (Zhang et al., 2014). Grandparent care is not the same as parental care (Wang and Bansak, 2024), a pattern documented even where the grandparent's own resources rise sharply (Duflo, 2003). This literature treats who cares for the child as given, and asks what that care does to the child. Left-behind childhood is itself one of urbanization's costs, paid by a child who did not choose it. Whether the pension studied here adds to that cost, or eases it, is still an open question.

3 Data

According to the institutional background described above, whether the pension changed who cared for a child is closely tied to whether it eased a family's duty to support its elderly. We use three survey data to answer our question. The China Health and Retirement Longitudinal Study (CHARLS) provides the pension receipt and grandparental care that we want to study. The China Family Panel Studies (CFPS) captures the link between grandparent, adult children and their grandchild. The China Migrants Dynamic Survey (CMDS) provides supporting evidence on whether an adult parent migrates with a child, which supports the same story from the adult parent's side.

3.1 The China Health and Retirement Longitudinal Study (CHARLS)

The China Health and Retirement Longitudinal Study (CHARLS) interviews the same rural respondents in 2011, 2013, 2015, and 2018, within five years of age sixty on either side: about 21,400 person-wave records report whether a respondent received the pension, and about 20,700 report how much. CHARLS also records whether the respondent cared for a grandchild, in 17,233 person-wave records. Hence we could compare respondents just above and just below age sixty, since the pension arrived at sixty whether or not a respondent had ever paid into it.

3.2 The China Family Panel Studies (CFPS)

The China Family Panel Studies (CFPS) provides the link between the grandparent's household and the grandchild's care: whether a child lived with each parent, and whether a grandparent helped with homework. We use three waves, 2014, 2016, and 2018; we do not use the 2020 wave, when CFPS changed its interview mode during the pandemic. About 4,682 children report how many months they lived with their father, and about 4,694 report their mother. So we could compare children whose grandparent had just crossed age sixty to children whose grandparent had not, before and after the reform.

3.3 The China Migrants Dynamic Survey (CMDS)

The China Migrants Dynamic Survey (CMDS) provides supporting evidence from the parent's side: whether a migrant reports difficulty caring for children or elderly parents left behind, and whether a migrant brought a child along on the first move. We use the 2017 wave and link each migrant's home province to that province's pension benefit level, with destination-province fixed effects and standard errors clustered by the thirty-one home provinces. About 66,895 migrants report the caregiving burden, and about 97,814 report whether they migrated with a child. Because home province is not randomly assigned, this evidence is associational, not causal. So we could ask whether a more generous pension at home lines up with less caregiving burden and more migration with children.

Table 1 summarizes the three samples. The CHARLS sample follows 21,433 grandparents near the pension cutoff. The CFPS sample follows 4,711 children living with those grandparents. The CMDS sample includes 66,895 migrants asked about caregiving burden and 97,814 asked whether they migrated with a child.

4 Empirical Analysis: The Age-Sixty Cutoff

According to the institutional background described above, whether the pension changed who cared for a child is closely tied to whether it eased a family's duty to support its elderly. To exploit the age-sixty cutoff, we compare households just above and just below it, before and after the 2016 reform, in our sample. Hence the change in behavior that appears only after 2016, and only at the

Table 1. Summary Statistics

	Observations	Mean	Median	Std. Dev.
<i>Panel A: CHARLS (age-sixty sample, 2011–2018)</i>				
Receives the pension	21,433	0.367	—	0.482
Pension income (yuan/year)	20,700	612.1	0	2,101.9
Cares for a grandchild	17,233	0.577	—	0.494
Co-resides with an adult child	21,206	0.498	—	0.500
Working	21,331	0.748	—	0.434
<i>Panel B: CFPS (child sample, 2016 and 2018)</i>				
Months with father (of 12)	4,683	6.68	6	4.95
Months with mother (of 12)	4,695	8.13	12	4.92
Child age	4,710	6.98	7	4.00
Child is female	4,711	0.471	—	0.499
Father absent all year	4,711	0.116	—	0.320
Mother absent all year	4,711	0.109	—	0.312
Grandparent is a grandmother	4,711	0.517	—	0.500
Grandparent's years of schooling	4,499	4.89	6	4.57
Grandparent past sixty	4,711	0.583	—	0.493
<i>Panel C: CMDS (migrant sample, 2017)</i>				
Reports child-care burden	66,895	0.213	—	0.409
Reports eldercare burden	66,895	0.708	—	0.454
Migrated with a child, first move	97,814	0.277	—	0.447

Notes: Panel A: CHARLS, rural respondents aged 55–65, 2011–2018. Panel B: CFPS, children with a rural older relative aged 55–65, 2016 and 2018. Panel C: CMDS, 2017 migrants linked to home-province pension benefits. Unless noted, variables are indicators equal to one for the condition named.

cutoff, identifies the effect of the pension.

4.1 The RD-DID Specification: the effect on Grandparents

Our estimating equation is

$$Y_{igt} = \alpha + \beta (\text{Elig}_g \times \text{Post}_t) + \gamma_1 \text{Elig}_g + \gamma_2 \text{Post}_t + f(a_g) + X'_{it} \delta + \varepsilon_{igt} \quad (1)$$

Y_{igt} is the outcome for person i in grandparent g 's family at wave t . a_g is the grandparent's age minus sixty. Elig_g equals one once a_g reaches zero, and zero before. Post_t equals one for waves at or after 2016, and zero before. $f(a_g)$ is a running trend in age, fit separately on each side of the cutoff and interacted with Post_t , so a trend already present before the reform does not get counted

as the reform's effect. X_{it} holds covariates measured before the reform. A covariate measured after 2016 could itself have changed because of the reform, and controlling for it would hide part of the effect we want to measure. β is the number we want: how much the outcome moved for people just past sixty, once the reform made sixty a pension age. CHARLS and CFPS share this equation, but not its details: the variables in X_{it} , how standard errors are clustered, and the sample window differ between the two surveys; we report each below.

For CHARLS, X_{it} holds the respondent's gender and schooling category. Standard errors are clustered by individual. We use the four waves 2011, 2013, 2015, and 2018, and keep respondents within five years of age sixty on either side.

For CFPS, X_{it} holds the child's age and gender. Standard errors use the HC1 correction, which does not require the error variance to be the same for every observation. The main estimates compare 2016 to 2018; 2014 to 2016 serves as a placebo before the reform. We keep children within five years of age sixty on either side.

4.2 The Effect on Grandsons

For results split by the child's gender, we add one term to equation (1):

$$Y_{igt} = \alpha + \beta_1 (\text{Elig}_g \times \text{Post}_t) + \beta_2 (\text{Elig}_g \times \text{Post}_t \times \text{Boy}_i) + \gamma_1 \text{Elig}_g + \gamma_2 \text{Post}_t + f(a_g) + X'_{it} \delta + \varepsilon_{igt} \quad (2)$$

Boy_i equals one if child i is a boy, and zero if a girl. The running trend is interacted with Boy_i as well as with Elig_g and Post_t : $f(a_g)$, $f(a_g) \times \text{Elig}_g$, $f(a_g) \times \text{Post}_t$, and $f(a_g) \times \text{Elig}_g \times \text{Post}_t$ each appear separately for boys and girls. β_1 is the same object as β in equation (1), estimated on the pooled sample of girls and boys. β_2 tells us whether the reform moved the outcome more for boys than for girls. We report boys and girls in separate columns rather than β_2 alone, because the split shows the size of the effect for each group, not just the difference between them. The p-value on β_2 appears in the table notes.

4.3 Evidence from the Parent’s Side

As supporting evidence, we ask whether a more generous pension at home lines up with a migrant’s caregiving burden and migration choices, using the 2017 CMDS wave:

$$M_{ipd} = \theta \ln(\text{Benefit}_p) + \phi g_p + Z'_i \lambda + \mu_d + u_{ipd} \quad (3)$$

M_{ipd} is the outcome for migrant i from home province p to destination province d : whether the migrant reports difficulty caring for children, whether the migrant reports difficulty caring for elderly parents left behind, or whether the migrant brought a child along on the first move. Benefit_p is the 2017 pension benefit level in the migrant’s home province. No single national source reports this level for all thirty-one provinces, so we collected it by hand from each province’s own government bulletin (see Appendix XXX). g_p is the log of the home province’s GDP, added so the pension is not standing in for the wealth of the province that pays it. Z_i holds the migrant’s age, gender, education, and family size. μ_d is a destination-province fixed effect, so we compare migrants who ended up in the same place. Standard errors are clustered by the thirty-one home provinces. Home province is not randomly assigned, so θ is associational, not causal.

4.4 Testing the Cutoff

For every CFPS result, we also report a randomization-inference p-value. We re-estimate the same equation at twelve fake cutoffs, ages fifty-two through fifty-seven and sixty-three through sixty-eight, as if each were the pension’s true threshold. We drop ages fifty-eight through sixty-two from this set, since a fake cutoff that close to sixty would fall inside the true discontinuity’s own window. The randomization-inference p-value ranks the true cutoff’s t-statistic against the twelve fake ones: $p_{RI} = (\#\{|t_{fake}| \geq |t_{true}|\} + 1)/13$. Within each family of results, we correct for testing many outcomes at once using the Benjamini-Hochberg procedure. A bandwidth of two years is too narrow to fit a local quadratic trend without the standard errors absorbing the effect we want to measure, so we check the running trend’s shape at a wider bandwidth instead, five and seven years on either side of the cutoff.

5 Three Generations

5.1 The Grandparents

The reform's first effect falls on the pension itself. Eligible respondents are 18.2 percentage points more likely to receive a pension after 2016, against a baseline rate of 36.7 percent (Table 2, column 1). Pension income rises by 561 yuan a year, on a base of 612 yuan (Table 2, column 2). Grandparents do not use this income for more leisure. After 2016, eligible grandparents are 7.7 percentage points more likely to care for a grandchild, against a baseline of 57.7 percent (Table 2, column 3), consistent with a pension letting adult children work and live away from home while grandparents cover the child care left behind. The grandmothers who cover this care do not work less: they are 12.1 percentage points more likely to be working themselves, against a baseline of 67.7 percent (Table 2, column 4).

This care increase does not fall evenly on grandmothers and grandfathers, though the two groups are not statistically distinguishable on caregiving alone. Grandmothers are 10.6 percentage points more likely to care for a grandchild after 2016, against a baseline of 57.9 percent, and this rise is significant on its own (Table 3, column 1). Grandfathers rise by 4.9 percentage points, against a baseline of 57.6 percent, and this rise is not significant on its own (Table 3, column 2); the 5.2 percentage point gap between them is not significant either ($p=0.136$; Table 3, column 3). Their labor supply is easier to tell apart. Grandmothers are 12.1 percentage points more likely to be working, against a baseline of 67.7 percent (Table 3, column 1); grandfathers fall by an insignificant 4.1 percentage points, against a baseline of 82.5 percent (Table 3, column 2). This 16.2 percentage point gap is statistically significant ($p=0.004$; Table 3, column 3). Grandmothers do more caregiving and more work; grandfathers do neither.

Grandparents do not only cover more hours; they start a task their own schooling did not prepare them for. Conditional on a child receiving help with schoolwork, a grandparent is 37.9 percentage points more likely to provide it, against a baseline of 23.2 percent (Table 2, column 10). The grandparents taking on this role average only 4.9 years of schooling. Wang and Bansak (2024) find that grandparent caregiving lowers grandchildren's test scores, and that this effect concentrates among grandparents with less schooling. The grandparents doing this work may not teach as well as the parents they replace.

Table 2. Effects of Pension Eligibility at the Age-Sixty Cutoff

	(1) Receives pension	(2) Pension income (yuan/year)	(3) Cares for a grandchild	(4) Grandmother is working	
<i>Panel A. Pension receipt and grandparent outcomes</i>					
Eligible × post-2016	.182*** (.026)	560.9*** (177.8)	.077** (.034)	.121*** (.044)	
Mean of dependent variable	.368	612.1	.577	.677	
Observations	21,433	20,700	17,233	11,011	
	(5) Months co-resident with father	(6) Months co-resident with mother	(7) Either parent left the county		
<i>Panel B. Parental migration and co-residence</i>					
Eligible × post-2016	−1.389** (.596)	−1.609*** (.586)	.011 (.073)		
Mean of dependent variable	6.68	8.12	.330		
Observations	4,682	4,694	2,887		
	(8) Grandparent provides night care	(9) Parent provides night care	(10) Grandparent tutors child conditional on help	(11) Receives any tutoring	(12) Arguments with parents
<i>Panel C. Child care, tutoring, and parent–child conflict</i>					
Eligible × post-2016	.018 (.060)	.004 (.060)	.380*** (.140)	−.181** (.080)	1.631*** (.553)
Mean of dependent variable	.393	.515	.232	.409	.968
Observations	4,593	4,593	1,018	3,156	1,184

Notes: Each column reports a separate regression. The coefficient of interest is the interaction between pension eligibility at age sixty and the post-2016 period. The estimation sample consists of rural respondents or children within five years of age sixty on either side of the cutoff.

Columns (1)–(4) use CHARLS waves 2011, 2013, 2015, and 2018. The specifications control for respondent gender and schooling category, and standard errors are clustered at the individual level. Column (4) is estimated among grandmothers.

Columns (5)–(12) use CFPS waves 2016 and 2018. The specifications control for child age and gender and report heteroskedasticity-robust HC1 standard errors. The dependent variables in columns (5) and (6) are measured in months per year. Column (10) is conditional on the child receiving help with schoolwork. Column (11) captures the extensive margin of receiving any tutoring.

Specifications that additionally control for grandparent characteristics, alternative bandwidths, alternative tutoring definitions, and exploratory gender-specific estimates are reported in the appendix. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 3. Heterogeneity in the Effects of Pension Eligibility

	(1)	(2)	(3)
	Grandmothers	Grandfathers	Difference (grandmother – grandfather)
<i>Panel A. Grandparent outcomes by grandparent gender</i>			
Cares for a grandchild	0.106** (0.048)	0.049 (0.050)	0.052 (0.035)
Mean of dependent variable	0.579	0.576	0.577
Observations	9,021	8,212	17,233
Currently working	0.121*** (0.044)	–0.041 (0.035)	0.162*** (0.057)
Mean of dependent variable	0.677	0.825	0.748
Observations	11,011	10,320	21,331
	Father's months co-resident		
	(4)	(5)	(6)
	Boys	Girls	Difference (girls – boys)
<i>Panel B. Parental co-residence by child gender</i>			
Eligible × post-2016	–2.009** (0.821)	–0.721 (0.867)	1.289 (1.194)
Mean of dependent variable	6.70	6.65	6.67
Observations	2,478	2,204	4,682
	Mother's months co-resident		
	(7)	(8)	(9)
	Boys	Girls	Difference (girls – boys)
Eligible × post-2016	–2.100*** (0.801)	–1.056 (0.859)	1.049 (1.174)
Mean of dependent variable	8.00	8.26	8.12
Observations	2,484	2,210	4,694
	(10)	(11)	(12)
	Full sample	Boys	Girls
<i>Panel C. Grandparent tutoring by child gender</i>			
Eligible × post-2016	0.379*** (0.140)	0.758*** (0.179)	0.043 (0.198)
Mean of dependent variable	0.232	0.268	0.188
Observations	1,018	560	458
	(14)	(15)	(16)
	Grandmothers	Grandfathers	Difference (grandmother – grandfather)
<i>Panel D. Grandparent tutoring by grandparent gender</i>			
Eligible × post-2016	0.475** (0.189)	0.307 (0.202)	0.168 (0.277)
Mean of dependent variable	0.197	0.272	0.232
Observations	543	475	1,018
	(17)	(18)	(19)
	Full sample	Boys	Girls
<i>Panel E. Arguments with parents by child gender</i>			
Eligible × post-2016	1.631*** (0.553)	1.289** (0.604)	1.922** (0.925)
Mean of dependent variable	0.968	0.862	1.080
Observations	1,184	610	574
	(20)	Difference (girls – boys)	

Notes: Each entry reports the coefficient on eligible × post-2016; standard errors are in parentheses. Difference columns come from pooled regressions containing the corresponding triple interaction. Panel A uses rural CHARLS respondents aged 55–65 in waves 2011, 2013, 2015, and 2018, controls for schooling, and clusters standard errors by individual. Currently working equals one if the respondent reports working at the time of the survey; its pooled difference specification fully interacts the controls with grandparent gender. Panels B–E use CFPS waves 2016 and 2018, control for child age and, where applicable, child gender, and report HC1 robust standard errors. Child-gender differences are girls minus boys; grandparent-gender differences are grandmothers minus grandfathers. Panels C and D use the two-year bandwidth and condition on the child receiving help with schoolwork. Exact difference-test p -values are 0.136 for grandchild care and 0.004 for currently working in Panel A; 0.280 for father's months and 0.372 for mother's months in Panel B; 0.007 in Panel C; 0.544 in Panel D; and 0.567 in Panel E. The full-sample result in Panel E passes the twelve-placebo-cutoff test, with rank $p = 0.077$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 The Parents

The pension moves the intensive margin of parental absence, not the extensive margin. After 2016, eligible parents' children spend 1.4 fewer months a year with their father, against a baseline of 6.67 months (Table 2, column 5). They spend 1.6 fewer months with their mother, against a baseline of 8.12 months (Table 2, column 6). Whether a parent leaves the home county at all does not move: the estimate is 1.1 percentage points and is not statistically significant, against a baseline rate of 33.0 percent (Table 2, column 7). Parents who already work away from home spend even more time away; the pension does not send new parents away for the first time.

Evidence from migrants themselves points the same way. The China Migrants Dynamic Survey follows people who have already left home, matched to each origin province's average pension benefit. Migrants from a province with twice the average pension benefit are 2.9 percentage points less likely to report difficulty arranging child care back home, against a baseline of 21.3 percent ($p=0.004$). They are also 3.8 percentage points more likely to have brought a child along on their very first move, against a baseline of 27.7 percent ($p=0.006$). This evidence is associational, not causal: migrants sort into destinations for many reasons besides the local pension. Gai et al. (2025) find that the same reform freed adult children to leave the farm in the first place. That effect is about whether a parent migrates at all. This one is not: our own results already show that a parent's chance of leaving the home county does not move. What lines up with the pension here is a choice made after migrating: not whether the parent goes, but whether the child comes too. Where the pension is more generous, parents worry less about child care and more often bring the child along.

5.3 The Left-Behind Children

Not everything changed hands. We test six ordinary parts of a child's day: who provides night-time supervision, who takes the child to and from school, whether the child boards away from home during the week, how often the child hears a bedtime story, whether the child is enrolled in a tuition class, and what time the child goes to bed. On every one of these, the estimate is close to zero, and none approaches conventional significance (Table 4). This is not a small sample failing to detect an effect: night-time supervision alone is estimated on 4,593 children. Whoever supervised a child at night, walked the child to and from school, or put the child to bed before a grandparent turned

sixty still did the same after.

Table 4. Daily Care Inputs at the Age-Sixty Cutoff

Outcome	Estimate	SE	Mean	N	RI rank- <i>p</i>
Night-time supervision: grandparent	0.018	(0.060)	0.393	4,593	0.385
Night-time supervision: parent	0.004	(0.060)	0.515	4,593	0.385
School run: grandparent	0.156	(0.114)	0.429	1,144	0.385
School run: parent	-0.103	(0.112)	0.400	1,144	0.923
Boards away during the school week	0.084	(0.118)	0.477	1,009	1.000
How often the child hears a bedtime story	-0.184	(0.359)	2.682	1,123	0.385
Enrolled in tuition class	-0.041	(0.039)	0.096	4,050	0.154
Bedtime (minutes since midnight)	0.648	(6.028)	1,273.2	4,033	0.615

Notes: Each row reports the coefficient on pension eligibility at age sixty interacted with the post-2016 period. Estimates use the CFPS child sample in 2016 and 2018 with the paper's default five-year bandwidth, controlling for child age and gender and reporting HC1 robust standard errors. The bedtime variable is minutes elapsed since the most recent midnight; the sample mean of 1,273 minutes is about 9:13 pm. The RI rank-*p* column uses the archived twelve-placebo-cutoff test with a two-year bandwidth. Conventional main-specification *p*-values are all 0.172 or larger. The Benjamini-Hochberg multiple-testing correction gives the same adjusted value for every row, $q = 0.941$.

Boys lost more time with their parents than girls did, in point estimate. After 2016, boys spent 2.0 fewer months a year with their father and 2.1 fewer with their mother; for girls the same estimates were 0.7 and 1.1 months. The difference between boys and girls is not itself significant (Table 3, Panel B). The same asymmetry shows up in who helps with homework. Among children who already had someone helping, the chance that the helper was a grandparent rose 75.8 percentage points for boys, against a mean of 26.8 percent, and 4.3 points for girls, against a mean of 18.8 percent, at a two-year bandwidth (Table 3, Panel C). This gap, 71.6 points, is significant (SE 26.7). It does not survive a wider window: at a five-year bandwidth, the split is 9.0 points for boys and -3.1 points for girls, neither significant. We report the concentration on boys as a narrow-window result, not a general one. Both point the same way, toward the son.

This pattern is not unique to this setting. Bertrand and Pan (2013) find that home environment explains much of the gender gap in disruptive behavior among American children: boys respond more than girls to weaker parental supervision and family instability. Autor et al. (2019) reach the same conclusion from a different source, the near-universe of births in one American state: family disadvantage widens the gender gap in behavior and schooling, and boys bear more of it than girls. This paper's own pattern fits that literature: more months without a parent, and a grandparent with less schooling helping with homework, are exactly the kind of change in a child's environment this literature ties to a wider gender gap.

Conflict is the exception to that pattern. Parent-child arguments also rose: children report 1.631

more arguments a year with their parents after 2016, against a mean of 0.968 (Table 2, Panel C, column 12). This estimate also passes a stricter test: the randomization-inference p-value, based on twelve placebo cutoffs, is 0.077. But this result does not split by the child's gender the way tutoring does. Boys report 1.289 more arguments and girls report 1.922 more, against means of 0.862 and 1.080 (Table 3, Panel E). The gap between them is not significant (SE 1.105). More conflict at home follows the pension, but it does not settle on the son.

6 Conclusion

When rural workers left for the cities to earn more for the families they left behind, two questions followed: who would ease the burden of supporting their parents, and who would raise their children. The pension partially answered the first question. This paper has traced what that answer did to the second. At sixty, a grandparent began to draw the benefit. The household let the adult child stay away longer: children spent 1.4 fewer months a year with their father and 1.6 fewer with their mother, while the chance that a parent left home at all did not move. Grandmothers did more of both: more grandchild care and more work; grandfathers did neither. Help with homework, the one task that had waited for a parent, passed to grandparents averaging 4.9 years of schooling, less than a completed primary education. The grandmothers among them average only 2.9 years, and at least half never attended school. What the son once owed his parents did not disappear when the state began to pay. It changed shape, from money into time, and it traveled down a generation. Support for the old moved from the son to the state; the cost of the move landed on the son's own son.

The question this paper ends on is not particular to China. Wherever the state takes over the support of the old from the family, the benefit enters a household, and its effects divide among the members. Duflo (2003) followed South Africa's pension inside the household and found good news: paid to a grandmother, it made her granddaughters measurably healthier. Gai et al. (2025) evaluate China's rural pension at the household level, treating the household as one decision maker, and find that the reform made it better off. Nothing in this paper contradicts that total; this paper is about its division. Part of the household's gain was paid for by its youngest member, in months with a mother and father. An account kept at the household level shows the gain and hides the payment.

The division was not bargained by everyone it covered. A household that sends a parent to

the city strikes a bargain: the parent's wage against the parent's presence. Sen (2001) calls such bargains cooperative conflict: the family gains from striking them, and the terms decide which member gains how much. The terms here were set by the adults. The children could not choose between a parent at home and a parent earning city wages, and the terms their households settled on gave them less of both parents. On the question of which children paid most, the evidence is thinner. Boys lost more months with each parent than girls did in point estimate, but the difference is not significant. The homework handoff does concentrate on boys, at the two-year bandwidth used for that outcome, and not at wider ones. And whether these childhoods carry a cost into adulthood, the data end too early to say.

Part of the bargain's terms was set by policy. The pension pays money, but money was not what kept the children in the village. A child's *hukou* stays there, and for the cohorts this paper follows, a village *hukou* made a seat in a city public school costly and uncertain, though rarely impossible: admission ran through permits, insurance records, points, and the supply of places, and it was hardest where the wages were highest. In the CMDS, migrants from provinces with more generous pensions are 3.8 percentage points more likely to have brought a child along on their first move. The pattern is associational, but it fits parents who bring the child when the terms allow it. The terms have been loosening city by city: settlement limits were lifted in towns and small cities in 2014 (State Council of the People's Republic of China, 2014), then in cities of up to three million urban residents by 2019 (National Development and Reform Commission of the People's Republic of China, 2019), while the largest cities still ration settlement by points. In income, these households gained: the pension arrived, and the city wage followed the parent who left. In the terms of Sen (2001), low income is not the only deprivation; there is also capability deprivation, a narrowing of what a person can do and be. The months these children lost with their mother and father are that second kind of loss: no income statistic records them, and the *hukou* reforms that came later cannot give them back.

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